

Challenging Questions-Solution



(1,4), (2,15), (3,30), (4,49), (5,72), (6,99)

x	y	Finite or First Differences Δy	Second Differences $\Delta^2 y$
1	4		
2	15	11	4
3	30	15	4
4	49	19	4
5	72	23	4
6	99	27	

$$f(x) = 2x^2 + 5x - 3$$



(1, -34), (2,-42), (3,-38), (4,-16), (5,30), (6,106)

x	y	Finite or First Differences Δy	Second Differences $\Delta^2 y$	Third Differences $\Delta^3 y$
1	-34			
2	-42	-8	12	
3	-38	4	18	6
4	-16	22	24	6
5	30	46	30	6
6	106	76		

$$f(x) = x^3 - 15x - 20$$

Challenging Questions-Solution



(1,12), (2,-10), (3,-18), (4,0), (5,56), (6,162)

x	y	Finite or First Differences Δy	Second Differences $\Delta^2 y$	Third Differences $\Delta^3 y$
1	12			
2	-10	-22	14	
3	-18	-8	26	12
4	0	18	38	12
5	56	56	50	12
6	162	106		

$$f(x) = 2x^3 - 5x^2 - 21x + 36$$



(1,-2), (2,-4), (3,-6), (4,-8), (5,14), (6,108), (7,346)

x	y	Finite or First Differences Δy	Second Differences $\Delta^2 y$	Third Differences $\Delta^3 y$	Fourth Differences $\Delta^4 y$
1	-2				
2	-4	-2	0		
3	-6	-2	0	0	
4	-8	-2	24	24	24
5	14	22	72	48	24
6	108	94	144	72	
7	346	238			

Challenging Questions-Solution



$(1,-2), (2,-4), (3,-6), (4,-8), (5,14), (6,108), (7,346)$

$$f(x) = x^4 + bx^3 + cx^2 + dx + e$$

$$f(1) = 1 + b + c + d + e = -2$$

$$b + c + d + e = -3$$

$$f(2) = 16 + 8b + 4c + 2d + e = -4$$

$$8b + 4c + 2d + e = -20$$

$$f(3) = 81 + 27b + 9c + 3d + e = -6$$

$$27b + 9c + 3d + e = -87$$

$$f(4) = 256 + 64b + 16c + 4d + e = -8$$

$$64b + 16c + 4d + e = -264$$

$$e = -3 - b - c - d$$

$$7b + 3c + d = -17$$

$$b = -10$$

$$26b + 8c + 2d = -84$$

$$c = 35$$

$$63b + 14c + 3d = -261$$

$$d = -52$$

$$e = 24$$



$$f(x) = x^4 - 10x^3 + 35x^2 - 52x + 24$$



2. The first three square pyramidal numbers are 1, 5, and 14, as shown in the diagram. Find the next three pyramidal numbers and determine the equation of a polynomial function that gives the x^{th} square pyramidal number.

$$f(1) = 1$$

$$f(2) = 5$$

$$f(3) = 14$$



Challenging Questions-Solution



x	f(x)	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$\Delta^4 f(x)$
1	1				
2	5	4			
3	14	9	5		
4	30	16	7	2	
5	55	25	9	2	
6	91	36	11	2	



$$\Delta^3 f(x) = an!$$

$$2 = a(3!)$$

$$2 = 6a$$

$$a = \frac{1}{3}$$

$$f(1) = 1$$

$$f(2) = 5$$

$$f(3) = 14$$

$$f(x) = ax^3 + bx^2 + cx + d$$

$$f(x) = \frac{1}{3}x^3 + bx^2 + cx + d$$

$$1 = \frac{1}{3} + b + c + d$$

$$5 = \frac{8}{3} + 4b + 2c + d$$

$$14 = 9 + 9b + 3c + d$$

$$b = \frac{1}{2}, c = \frac{1}{6}, d = 0$$



$$\therefore f(x) = \frac{1}{3}x^3 + \frac{1}{2}x^2 + \frac{1}{6}x$$

$$f(x) = \frac{1}{6}(2x^3 + 3x^2 + x)$$

$$= \frac{1}{6}x(2x^2 + 3x + 1)$$

$$= \frac{1}{6}x(x+1)(2x+1)$$

which is the formula for the sum $1^2+2^2+3^2+4^2+\dots+x^2$.